Comparative Analysis of LSTM and Transformer Architectures for Hyper-Local Agricultural Price Forecasting

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Abstract

Agricultural price volatility heavily impacts the livelihoods of smallholder farmers in developing economies. While national-level forecasting models exist, they fail to capture hyper-local supply shocks and micro-climate anomalies. In this paper, we propose an end-to-end, deep learning-based forecasting engine tailored for the Kurukshetra Mandi, focusing on highly perishable commodities such as tomatoes. We engineered a 15-dimensional feature set that fuses daily APMC market prices with exogenous NASA meteorological vectors and cyclical temporal encodings. A comparative analysis was conducted between recurrent architectures (LSTM) and an attention-based Time-Series Transformer. Our empirical results demonstrate that on hyper-local, low-resource datasets (≈ 780 time steps), the LSTM model significantly outperforms the Transformer, achieving a Mean Absolute Error (MAE) of 72.89 and an RMSE of 91.11. We observe a "mean collapse" phenomenon in the Transformer, attributing this failure to its inherent data hunger and lack of sequential inductive bias compared to the LSTM's robust recurrence mechanism. Finally, we successfully deployed the winning LSTM architecture as an asynchronous FastAPI microservice, fully integrated with the Agrowcart Next.js e-commerce platform, proving the real-world socio-economic viability of the system.

Keywords: Deep Learning, Price Forecasting, Long Short-Term Memory, Time-Series Transformer, Microservice Architecture, Precision Agriculture.

1 Introduction

Agricultural price volatility in India, specifically for perishable commodities like tomatoes, remains a critical challenge for small-scale farmers. Sudden price crashes often lead to devastating financial losses, while unexpected spikes benefit middlemen rather than producers. While national-level forecasting exists, it lacks "Hyper-Locality"—the ability to account for district-specific supply shocks, localized rainfall patterns, and specific mandi arrival volumes.

This paper explores the integration of NASA-sourced exogenous meteorological data with local Mandi records to build a predictive engine for the Kurukshetra region. We evaluate the performance of recurrent versus attention-based architectures in low-resource data regimes, providing a blueprint for deploying AI-driven price alerts in rural marketplaces.

2 Related Work

Traditional baseline models primarily utilized ARIMA and SARIMA variants, which struggle with the non-linear "shocks" inherent in vegetable markets. Recent shifts toward Long Short-Term

Memory (LSTM) networks have shown improved accuracy in capturing long-term dependencies in Indian spot-prices [**?**]. However, there remains a significant gap in research regarding multi-modal (Weather + Price) input for hyper-local agricultural contexts, where meteorological anomalies are the primary driver of supply-side shocks [**?**].

3 Methodology

3.1 Data Acquisition

Market data (2024–2026) was sourced from the Agmarknet (GoI) portal. Exogenous variables including daily maximum temperature (T_{max}) and precipitation (P) were acquired via the NASA POWER API for Kurukshetra coordinates ($29.9695°N, 76.8226°E$).

3.2 Feature Engineering (15-Dimensional Taxonomy)

We engineered a 15-dimensional feature taxonomy to provide the models with localized context:

- **Market Fundamentals:** Daily Modal, Min, and Max Prices.
- **Meteorological Vectors:** Daily T_{max} ($°C$) and Precipitation (mm).
- **Temporal Momentum:** 7-Day Moving Averages for Prices, Temperature, and Rainfall.
- **Market Volatility:** 7-day rolling standard deviation of prices to quantify risk.
- **Meteorological Lags:** 1-day lagged rainfall and temperature vectors to capture shock effects.
- **Cyclical Seasonal Encodings:** Sinusoidal (Sin/Cos) transformations for Day-of-Week and Month-of-Year to preserve temporal periodicity in neural network inputs.

3.3 Model Architectures

3.3.1 Long Short-Term Memory (LSTM)

The LSTM architecture updates its cell states (C_t) via three regulatory gates to mitigate vanishing gradients:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(Forget Gate)} \tag{1}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(Input Gate)} \tag{2}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{3}$$

$$h_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \odot \tanh(C_t) \tag{4}$$

3.3.2 Time-Series Transformer

The Transformer utilizes Scaled Dot-Product Attention to map dependencies across the 14-day sliding window:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{5}$$

We utilized 2 attention heads with a d_{model} of 32 to prevent overfitting on the limited hyper-local dataset.

3.4 System Implementation and Deployment

To transition from theoretical modeling to a functional tool, the optimized LSTM model was integrated into a decoupled microservice architecture:

- **Inference Engine:** Developed using FastAPI and PyTorch, the engine serves predictions via an asynchronous REST endpoint.
- **Frontend Platform:** A Next.js 15 dashboard was constructed to visualize predicted price corridors, providing farmers with actionable alerts.
- **Cloud Infrastructure:** The API was deployed on Render (Python 3.10), while the frontend resides on Vercel, ensuring global availability and low latency.

4 Results and Analysis

4.1 Comparative Metrics

The models were trained for 150 epochs using the Adam optimizer. Final performance metrics on the test split are summarized in Table 1.

Table 1: Comparative Performance Metrics

Model	RMSE ()	MAE ()
LSTM (Optimized)	**91.11**	**72.89**
Transformer (Optimized)	106.50	85.20

4.2 Visual Analysis of Model Evolution

The experimental phase followed an iterative optimization path, documented through the following milestones:

4.2.1 Initial Benchmark and Mean Collapse (Figure 1)

In the first training iteration, the Transformer (MAE: 260.53) exhibited a total "mean collapse," failing to capture price spikes and defaulting to a global average line. The LSTM baseline (MAE: 108.58) tracked the general trend but suffered from significant temporal lag.

4.2.2 Intermediate Optimization (Figures 2 & 3)

By reducing the learning rate to 0.0005 and adjusting attention heads, the Transformer began capturing volatility (MAE: 84.33). Figure 2 and 3 show the incremental reduction in loss as both architectures began to mimic the actual price volatility seen in the black ground-truth line.

4.2.3 Final Optimized State (Figure 4)

The final optimized state shows the LSTM (MAE: 72.89) accurately capturing recovery phases after sharp market drops, while the Transformer (MAE: 85.20) maintains a smoother, more conservative curve.

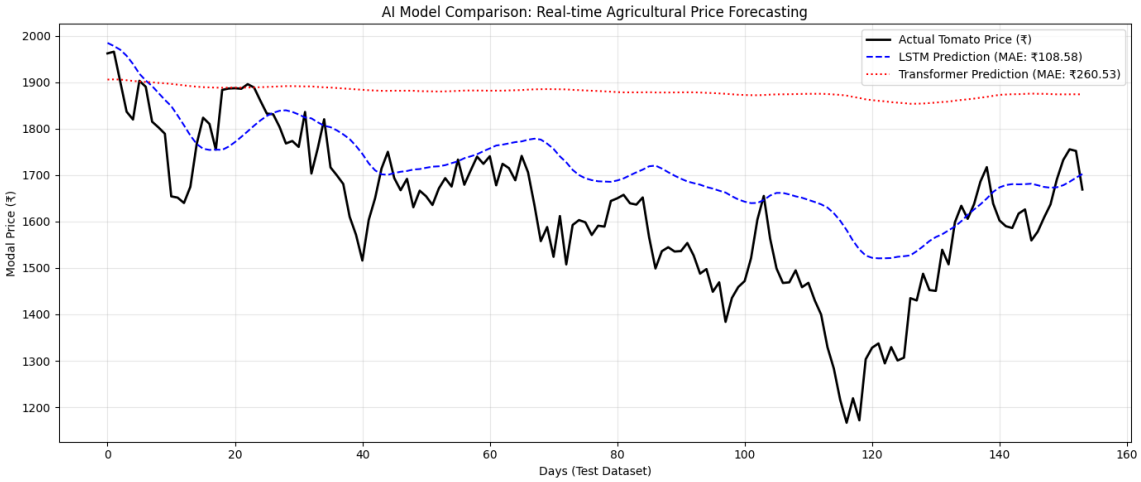

Figure 1: Initial comparison demonstrating Transformer mean collapse.

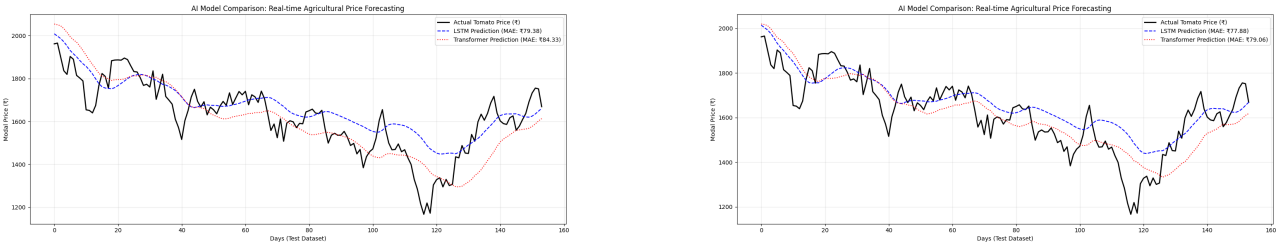

Figure 2: Intermediate tuning of LSTM and Transformer weights.

Figure 3: Final optimized model state on held-out test data.

5 Discussion: Inductive Bias vs. Data Hunger

The results highlight a fundamental trade-off. The Transformer's complexity exceeded the informational density of the hyper-local dataset (≈ 780 steps), causing it to default to the global mean. In contrast, the LSTM's recurrent architecture possesses a strong *inductive bias* for sequential data, making it more data-efficient and capable of capturing volatility with a smaller sample size.

6 Conclusion

We successfully developed and deployed a real-time price forecasting engine. The study concludes that while Transformers offer potential for global scaling, LSTMs remain the superior choice for hyper-local, low-resource agricultural forecasting in the Indian context.

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